INDUSTRIAL CONVEYOR SYSTEM and DRIVE MOTOR
FIELD SERVICE and MAINTENANCE MANUAL
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Equipment Class: Belt Conveyor with Electric Drive Motor
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SECTION 1: SYSTEM OVERVIEW

The industrial belt conveyor system transports material along a continuous rubber belt driven by an electric motor through a gearbox and drive pulley. The major components a field technician services are: the conveyor belt, the drive and tail pulleys, the carrying and return idler rollers, the drive motor, the gearbox (speed reducer), the drive belt or chain coupling, and the bearing assemblies that support the rotating shafts.

Safety: Always lock out and tag out the motor power before servicing any moving part. Wait for all rotation to stop. Never service a tensioned belt without first releasing belt tension at the take-up unit.

SECTION 2: CONVEYOR BELT

Component identity: The conveyor belt is the wide, continuous rubber band that carries material along the length of the conveyor frame.

Failure mode 2.1 — Belt mistracking (belt drifts to one side)

Symptoms: The belt runs off-center, rubs the frame, or its edge frays.
Cause: Misaligned idler rollers, off-square pulleys, or uneven belt tension.
Corrective steps:

1. Lock out motor power and let the belt stop.
2. Inspect idler rollers across the span; check that they sit square to the frame.
3. Adjust the tracking idlers in small increments, moving the leading-side roller forward in the direction you want the belt to move.
4. Verify both take-up tension bolts are set equally; uneven tension pulls the belt.
5. Run the belt empty and observe tracking for several full cycles before loading.

Failure mode 2.2 — Belt slipping on the drive pulley

Symptoms: Belt stalls under load while the drive pulley keeps turning; squealing.

Cause: Insufficient belt tension, worn pulley lagging, or material on the pulley face.

Corrective steps:

1. Lock out power. Inspect the drive pulley face for worn or glazed lagging.
2. Clean material buildup from the pulley surface.
3. Increase belt tension at the take-up unit until slipping stops, without over-tensioning (which overloads bearings).
4. Replace the pulley lagging if it is worn smooth.

Failure mode 2.3 — Belt surface damage (cuts, gouges, fraying)

Symptoms: Visible tears, gouges, frayed edges, or exposed carcass fabric.

Cause: Trapped foreign objects, impact at the load point, or edge mistracking.

Corrective steps:

1. Lock out power. Mark and inspect the damaged section.
2. For small cuts, apply a vulcanized or mechanical belt patch per kit instructions.
3. For damage spanning more than one third of belt width, replace the belt section.
4. Identify and remove the cause (sharp object, misaligned chute) before restart.

SECTION 3: DRIVE MOTOR (ELECTRIC)

Component identity: The electric drive motor is the cylindrical motor with a cooling fan cover and terminal box, coupled to the gearbox; it powers the conveyor.

Failure mode 3.1 — Motor overheating

Symptoms: Motor housing too hot to touch, burning smell, thermal trip.

Cause: Overload, blocked cooling fan, low voltage, or failing bearings.

Corrective steps:

1. Stop the motor and allow it to cool. Lock out power.
2. Clear debris from the fan cover and cooling fins so airflow is unobstructed.
3. Measure supply voltage at the terminal box; confirm it is within nameplate range.
4. Check the running current against the nameplate full-load amps; if over, reduce conveyor load or investigate a mechanical bind in the drivetrain.
5. If overheating persists with normal load and voltage, inspect motor bearings (Section 6) and have the windings tested for insulation resistance.

Failure mode 3.2 — Motor will not start (hums but no rotation)

Symptoms: Motor hums or buzzes when energized but the shaft does not turn.

Cause: Failed start capacitor, single-phasing (lost phase), or a seized load.

Corrective steps:

1. Lock out power. Try to rotate the shaft by hand; if seized, the fault is mechanical (gearbox or bearing) — see Sections 5 and 6.
2. If the shaft turns freely, check all three supply phases at the terminal box; a missing phase causes humming without rotation.
3. On single-phase motors, test and replace the start capacitor if faulty.
4. Inspect the contactor and overload relay for failure.

Failure mode 3.3 — Excessive motor vibration

Symptoms: The motor shakes, walks on its mounts, or produces a rhythmic knock.

Cause: Loose mounting bolts, shaft misalignment to the gearbox, or worn bearings.

Corrective steps:

1. Lock out power. Check and torque all motor foot mounting bolts.
2. Verify shaft-to-gearbox coupling alignment with a straightedge or dial gauge.
3. If vibration continues, inspect the motor bearings for play or roughness (Section 6).

SECTION 4: DRIVE AND TAIL PULLEYS

Component identity: Pulleys are the large drums at each end of the conveyor; the drive pulley (motor end) turns the belt, the tail pulley (far end) redirects it.

Failure mode 4.1 — Worn or damaged pulley lagging

Symptoms: Smooth, glazed, or chunked rubber coating on the pulley face; belt slip.

Cause: Normal wear, abrasive material, or chronic belt slip.

Corrective steps:

1. Lock out power and release belt tension.
2. Inspect the lagging across the full face for wear or missing sections.
3. Re-lag or replace the pulley if lagging is worn smooth or detached.
4. Confirm the pulley is square to the belt path after reinstallation.

Failure mode 4.2 — Pulley bearing failure

Symptoms: Grinding or rumbling from a pulley end; the pulley is hot or wobbles.

Cause: Contamination, loss of lubrication, or fatigue.

Corrective steps:

1. Lock out power. Rotate the pulley by hand and feel for roughness or play.
2. Replace the pillow-block bearing on the affected pulley shaft (see Section 6).
3. Re-check pulley alignment after bearing replacement.

SECTION 5: GEARBOX (SPEED REDUCER)

Component identity: The gearbox is the sealed cast housing between the motor and the drive pulley that reduces motor speed and increases torque.

Failure mode 5.1 — Oil leak from the gearbox

Symptoms: Oil pooling beneath the gearbox; low oil level in the sight glass.
Cause: Failed shaft seal, loose drain plug, or a cracked housing gasket.
Corrective steps:
1. Lock out power. Clean the housing to locate the leak source.
2. Tighten the drain and fill plugs if they are the source.
3. Replace the input or output shaft seal if oil leaks around a rotating shaft.
4. Replace the housing gasket if the leak is along the case joint.
5. Refill with the specified gear oil to the sight-glass line.

Failure mode 5.2 — Gearbox noise or overheating

Symptoms: Whining, grinding, or excessive heat from the gearbox.
Cause: Low oil level, worn gears, or a failing internal bearing.
Corrective steps:
1. Lock out power. Check the oil level and condition in the sight glass.
2. Top up or change the gear oil if low or contaminated (metal particles indicate internal wear).
3. If noise persists with correct oil, the gearbox requires internal inspection or replacement of worn gears and bearings.

SECTION 6: BEARINGS

Component identity: Bearings are the pillow-block or flange units that support each rotating shaft (motor, pulleys, idlers), allowing smooth rotation.

Failure mode 6.1 — Bearing noise and roughness

Symptoms: Grinding, rumbling, or squealing from a bearing; rough feel when rotated by hand; localized heat.

Cause: Loss of lubrication, contamination, or fatigue wear.

Corrective steps:

1. Lock out power. Rotate the shaft by hand and locate the rough or noisy bearing.
2. Re-grease sealed-for-life units only if a grease fitting is present; otherwise replace the bearing.
3. To replace: remove the shaft load, unbolt the pillow block, pull the worn bearing, fit the new bearing, and re-align the shaft.
4. Confirm free rotation and correct alignment before restoring power.

Failure mode 6.2 — Bearing overheating

Symptoms: Bearing housing too hot to touch shortly after start.

Cause: Over-greasing, over-tensioned belt, or misalignment.

Corrective steps:

1. Lock out power and let the bearing cool.
2. Check belt tension; an over-tensioned belt overloads the bearings — relieve it.
3. Verify shaft alignment.
4. If a sealed bearing runs hot with correct tension and alignment, replace it.

SECTION 7: IDLER ROLLERS

Component identity: Idlers are the small rollers along the conveyor frame that support the carrying (top) and return (bottom) runs of the belt.

Failure mode 7.1 — Seized or non-rotating idler

Symptoms: A roller that does not spin; flat-spotted roller; belt-edge wear nearby.

Cause: Bearing seizure from contamination or wear.

Corrective steps:

1. Lock out power. Spin each idler by hand to find seized units.
2. Replace any idler that does not rotate freely.
3. Confirm the replacement is square to the frame to maintain belt tracking.

Failure mode 7.2 — Material buildup on idlers

Symptoms: Caked material on rollers causing belt mistracking or vibration.

Cause: Carryback material adhering to rollers.

Corrective steps:

1. Lock out power. Scrape and clean material from the rollers.
2. Inspect and adjust the belt cleaner/scraper at the discharge pulley to reduce carryback.

SECTION 8: ROUTINE MAINTENANCE SCHEDULE

Daily: Visually inspect belt tracking and listen for unusual noise from motor, gearbox, and bearings.
Weekly: Check belt tension at the take-up; inspect pulley lagging; feel bearing housings for excess heat after a run.
Monthly: Check gearbox oil level; grease bearings fitted with grease points; torque-check motor and pillow-block mounting bolts; inspect idlers for free spin.
Quarterly: Verify motor running current against nameplate; check coupling/shaft alignment; inspect the full belt surface for damage.

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